02526 Mathematical Modeling

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Lecture 8: Optical Flow Feedback

April 7, 2026

Completion

- 37 groups, 35 finished (94.6%)
- almost all groups could get their implementations to work in controlled settings

Structure

- about 90% had a clearly recognizable academic structure covering introduction, method/model, results, discussion, and conclusion, even when some labels were phrased differently, such as “Problem and Background,” “Mathematical Model and Methodology,” or “Experimental Evaluation and Results”
- about 80% had separate blocks named precisely or close to ”Introduction,” ”Method,” ”Results,” ”Discussion,” and ”Conclusion”
- about 85% had a separate Discussion section
- for comparison, about 77% had clear structure in the Salami project, and approx. 86% had a separate Discussion section there

References

- about 70% of the reports include an explicit references/sources/literature section
- approx. 60% have correct referencing practice
- main problems:
  - reference list present but little or no actual citations in the text
  - not all bibliography entries cited in the text
  - insufficient/imprecise reference to online material
  - contents show no references section
  - references not organized in a separate section
- for comparison, about 11% included references to sources in the Salami project

Math and figures

- generally nice rendition of math formulas
- watch the margins when you have long math expressions
- generally nice use of figures
- main problems:
  - too small (almost illegible) figure fonts! Make figure fonts same as, or comparable to, the caption font or the main text font.
  - image aspect ratio (squeezed images)

Contents

- Introduction section:
  - motivate WHY the problem matters: OK
  - describe WHAT will be done: OK
  - state a clear GOAL: still needs some improvement (ca. 70%)
- Method section:
  - describe the model: OK
  - list assumptions: OK
  - define all symbols: needs improvement (ca. 70%)
  - provide enough detail for reproduction: needs improvement (ca. 60%)
- Discussion and Conclusion sections:
  - Are results interpreted, not just reported? YES! (ca. 90%)
- Proofread! Spelling and grammar are erroneous at times: “Mathmatical Modeling,” “Gassian filter,” ”the results was quite bad,” ”Salami Quality,” ”the golfer beings to draw the club back,” ”Rory Mcilroys / Rorys,” ”includes to much neighborhood information,” ”the Gaussian kernel were combined,” ...

Questions and discussion

- Feedback on the exercise itself
  - Was the exercise clear?
  - Too long?
  - Anything confusing?
- Feedback on the peer review process
  - Was the peer review useful?
  - Too time-consuming?
  - Suggestions?